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FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2025

TEST DATE: 12-01-2025

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

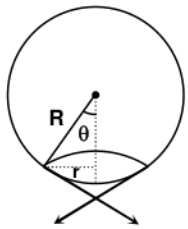
1. A

Sol. To detach

$$\rho_w\left(\frac{4}{3}\pi R^3 g\right) = T2\pi r\sin\theta$$

So, $\rho_w\left(\frac{4}{3}\pi R^3 g\right) = T2\pi r\frac{r}{R}$

So, $r = R^2\sqrt{\frac{2\rho_w g}{3T}}$



2. B

Sol. When switch is closed

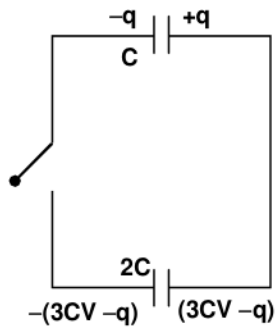
$$\frac{q}{C} = \frac{3CV - q}{2C}$$

So, $q = CV$

Initial energy, $U_i = \frac{C^2V^2}{2C} + \frac{(4CV)^2}{2(2C)}$

Final energy, $U_f = \frac{C^2V^2}{2C} + \frac{(2CV)^2}{2(2C)}$

So, energy lost = $U_i - U_f = 3CV^2$



3. A

Sol. $k_1 = \frac{hC}{\lambda} - \phi$

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$$3k_1 = \frac{2hC}{\lambda} - \phi$$

$$\text{So, } \frac{1}{3} = \frac{\frac{hC}{\lambda} - \phi}{\frac{2hC}{\lambda} - \phi}$$

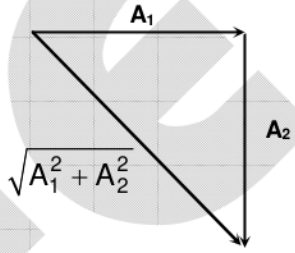
$$\text{So, } \phi = \frac{hC}{2\lambda}$$

4. A

Sol. From Phasor diagram

Because intensity I is proportional to square of amplitude

So, $I \propto (A_1^2 + A_2^2)$



5. A

Sol. When both S_1 and S_2 are open,

Current in the ammeter $i = \frac{1.5}{300+100+50} = \frac{1}{300}A$

When both S_1 and S_2 are closed,

$$i = \frac{1.5}{300 + \left(\frac{100R}{100+R}\right)}\left(\frac{R}{R+100}\right) = \frac{1.5R}{400R+30000}$$

So, according to the question,

$$\frac{1}{300} = \frac{1.5R}{400R+30000}$$

$\Rightarrow R = 600\ \Omega$

6. C

Sol. $$\frac{K(4)^2}{r^2(2)} = \frac{K(4-q)(4+q)}{r^2}$$

$$\text{So, } 8 = 16 - q^2$$

$\Rightarrow q = 2\sqrt{2}\ \mu C$

7. B

Sol. From Brewster's law

$\mu = \tan 60^\circ = \sqrt{3}$

8. B

Sol. $\frac{dy}{dx} = \frac{2a}{y} = \tan\theta$ (slope of the wire)

For no slipping $\tan\theta \le \mu_S$

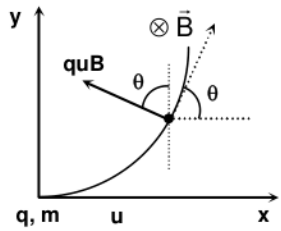
$$\text{So, } \frac{2a}{y_0} \le \mu_S$$

$$\text{So, } y_0 \ge \frac{2a}{\mu_S}$$

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9. B

Sol. Impulse momentum theorem along y-direction



$$\int quB\cos\theta dt = mu$$

$$\Rightarrow \quad q\int B_0 xu\cos\theta dt = mu$$

$$\Rightarrow \quad qB_0\int_0^{x_m} xdx = mu$$

$$\Rightarrow \quad qB_0\frac{x_m^2}{2} = mu$$

$$\Rightarrow x_m = \sqrt{\frac{2mu}{qB_0}}$$

10. A

Sol. $i_d = \frac{dQ}{dt} = \frac{12}{100}e^{-\frac{t}{\tau}}$

So, i_d after a time constant = $\frac{12000}{100e} = \frac{120}{e}$mA

11. D

Sol. $0.9A_0 = A_0e^{-c(5)}$

and $\alpha A_0 = A_0e^{-c(15)}$

So, $\alpha = (0.9)^3 = 0.729$

12. B

Sol. In the given circuit $R_{eq} = 1M\Omega$

and $C_{eq} = 4\,\mu F$

So, $V = (10)\left(1-e^{-\frac{t}{R_{eq}C_{eq}}}\right)$

So, $4 = 10\left(1-e^{-\frac{t}{4}}\right)$

So, $e^{-\frac{t}{4}} = \frac{3}{5}$

So, $t = 4\ell n\left(\frac{5}{3}\right)$

13. C

Sol. For a convex lens forming a real image

$$\frac{1}{v}-\frac{1}{-u}=\frac{1}{f}$$

So, if u = v, then

u = v = 2f

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14. B

Sol. In forced oscillation

If $F = F_0\cos(\omega t)$

Then $x = A\sin(\omega t - \phi)$, where $A = \dfrac{F_0}{m(\omega_0^2 - \omega^2)}$

15. B

Sol. Suppose length AP = x

So, $\dfrac{1}{3}\left(\dfrac{m}{\ell}x\right)x^2 = \dfrac{m\ell^2}{3} - \dfrac{mx^3}{3\ell}$

So, $x = \dfrac{\ell}{2^{1/3}}$

16. A

Sol. Heat required to melt the whole of ice is

$Q_1 = mS\Delta T + mL$

$= 400 \times 0.5 \times 20 + 400 \times 80 = 36000$ cal

The maximum heat released by steam when the whole steam is converted into water at 0°C

$Q_2 = mS\Delta T + mL$

$= 50 \times 1 \times 100 + 540 \times 50 = 32000$ cal

Because $Q_2 < Q_1$, so the whole ice will not melt. Hence the final temperature of the mixture is 0°C.

17. C

Sol. Thermal stress is given by $Y \propto \Delta\theta$

So, for stress to be same

$Y_1\alpha_1 = Y_2\alpha_2$

So, $\dfrac{Y_1}{Y_2} = \dfrac{\alpha_2}{\alpha_1} = \dfrac{3}{2}$

18. B

Sol. $Q = \dfrac{p^2}{2\times 4} + \dfrac{p^2}{2\times(220-4)}$

$5.5 = p^2\left(\dfrac{1}{8} + \dfrac{1}{432}\right) = p^2\left(\dfrac{440}{8\times 432}\right)$

So, $\dfrac{p^2}{2\times 4} = \dfrac{(5.5)(8\times 432)}{2\times 4\times 440} = 5.4$ MeV

19. C

Sol. From lens C, image is formed at 1 m along the direction of incident ray. So for the position of final image from the final system is

$\dfrac{1}{v} - \dfrac{1}{1} = \dfrac{1}{1}$

$\Rightarrow v = \dfrac{1}{2}$ m

So, magnification $m = \dfrac{1}{2}$

So, separation between both the images = $\dfrac{1}{2}\times\dfrac{1}{2}\times 2 = 0.5$ cm

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20. B

Sol. $X = \frac{C}{T}$

Or $\frac{X_1}{X_2} = \frac{273-173}{273-73} = \frac{100}{200}$

So, $X_2 = 2 \times 0.0060 = 0.0120$

SECTION – B

21. 770

Sol. For second pendulum

$$2 = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{\ell(1+\alpha\Delta T)}{g+a}}$$

So, $1+\alpha\Delta T = \frac{g+a}{g} = 1+\frac{a}{g}$

So, $a = g\alpha\Delta T = (10)(20\times10^{-4})(50) = 1\,m/s^2$

So, $N = m(g+a) = 770\ N$

22. 3

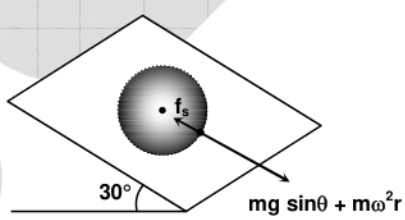
Sol. At the limiting situation

$f_S = mg\sin\theta + m\omega^2 r$

$mg\sin\theta + m\omega^2 r \le \mu_S mg\cos\theta$

So, $\mu_S \ge \tan\theta + \frac{\omega^2 r}{g\cos\theta}$

So, $\mu_S \ge \frac{1}{\sqrt{3}} + \frac{(10)^2\left(\frac{1}{10}\right)}{10\left(\frac{\sqrt{3}}{2}\right)} = \sqrt{3}$



23. 7

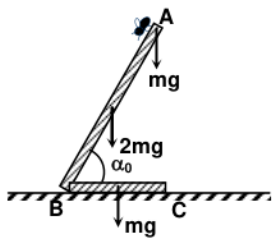
Sol. At the verge of toppling, normal applied by table will pass through point C. So, torque about point C,

$mg(2d\cos\alpha_0 - d) = 2mg(d - d\cos\alpha_0) + mg\frac{d}{2}$

$2\cos\alpha_0 - 1 = 2 - 2\cos\alpha_0 + \frac{1}{2}$

$4\cos\alpha_0 = \frac{7}{2}$

So, $\cos\alpha_0 = \frac{7}{8}$



24. 125

Sol. $\frac{dL}{dt} = \frac{dm}{dt}R(v_1 - v_2)$

$= (100)(0.5)(5-2.5) = 125\ J$

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25. 2

Sol. $X_C = \frac{V_C}{I} = 1000\ \Omega$

Current lags behind voltage, so box has an inductor

Power factor = 0.8 = $\frac{R}{Z} = \frac{800}{\sqrt{(800)^2 + (X_L - 1000)^2}}$

So, $X_L = 1600\ \Omega$

So, $L = \frac{1600}{2 \times \pi \times \frac{400}{\pi}} = 2$ Henry

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Chemistry

PART – B

SECTION – A

26. C

Sol. $K_2Cr_2O_7 + \text{cold. } 4H_2SO_4 \longrightarrow 2CrO_3 + 2KHSO_4 + 2H_2O$

$2K_2Cr_2O_7 + \text{warm } 8H_2SO_4 \xrightarrow{\Delta} 2K_2SO_4 + 2Cr_2(SO_4)_3 + 8H_2O + 3O_2$

27. C

Sol. $2F_2 + 2H_2O \longrightarrow 4HF(aq) + O_2(g)$

$3F_2 + 3H_2O \longrightarrow 6HF(aq) + O_3(g)$

28. B

Sol. $Ni^{+2} \rightarrow in\left[Ni(Br_2)(PPh_3)_2\right] \longrightarrow \left[NiBr_2(PPh_3)_2\right]$

Square planar	Tetrahedral
Red colour	Green coloured
Diamagnetic	Paramagnetic

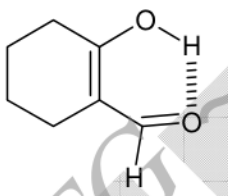
Tetrahedral splitting of Ni^{+2} in $NiBr_2(PPh_3)_2$ will have configuration $e^4t_2^4$, thus unpaired electrons.

29. D

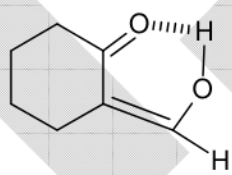
Sol. In (I) annulene systems, only peripheral bonds are counted but in (II) fused ring system only connected benzoic bonds are counted.

30. C

Sol.



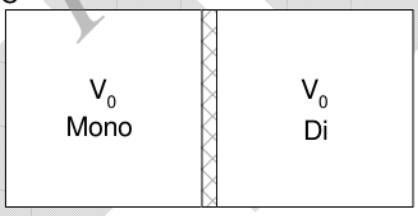
or



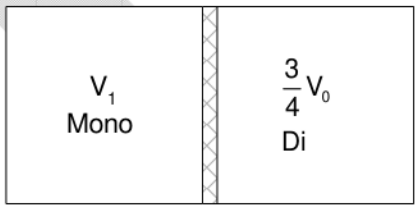
1, 2 diketones in enol form strong H-bonding with stable ring.

31. C

Sol.



Initial



Monoatomic : $P_1.V_0^{5/3} = P_2.V_1^{5/3}$

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Diatomic : $P_1.V_0^{7/5} = P_2.\left(\frac{3}{4}V_0\right)^{7/5}$

$\therefore \frac{V_1}{V_0} = \left(\frac{3}{4}\right)^{\frac{21}{25}}$

32. B

Sol.

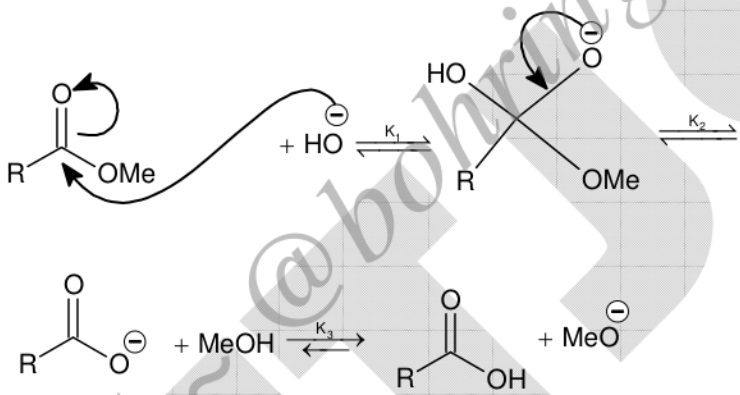
	I_2	+	I^-	$\rightleftharpoons$	I_3^-
	1		0.5		–
Volume = 1L	1 – 0.25		0.5 – 0.25		0.25

$K_c = \frac{0.25}{0.75 \times 0.25} = \frac{4}{3}$

$\therefore K_c$ for $\frac{1}{2}I_2 + \frac{1}{2}I^- \rightleftharpoons \frac{1}{2}I_3^-$ is $\sqrt{\frac{4}{3}} = 1.15$

33. C

Sol.



3 step mechanism where $K_1 \simeq K_2$ but K_3 is largely formed.

34. B

Sol. $\frac{1}{\lambda} = R_H.Z^2\left[\frac{1}{2^2} - \frac{1}{5^2}\right]$

$\frac{1}{108.5 \times 10^{-9}} = R_H.Z^2\left[\frac{25-4}{25 \times 4}\right]$

$Z^2 = \frac{1}{R_H}.\frac{1}{1085 \times 10^{-10}} \times \frac{25 \times 4}{21}$

$= \frac{916 \times 25 \times 4}{1085 \times 21}$

B. E. $= 13.6 \times \frac{Z^2}{n^2}$

B.E. $= \frac{+13.6 \times 916 \times 25 \times 4}{1085 \times 21}$

B.E. = 54.4 eV

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35. B

Sol.

	$A(g) \xrightarrow{K_1}$	$2B(g)$		$A(g) \xrightarrow{K2}$	$C(g)$
$t = 0$	1 atm	0	$t = 0$	1 atm	0
$t = 10$ min	$(1-x-y)$	$2x$	$t = 10$ min	$(1-x-y)$	y
$t = \infty$	$(1-a-b)$	$2a$	$t = \infty$	$(1-a-b)$	b
	≈ 0			≈ 0	

From question, $a + b = 1$ and $2a + b = 1.5$

$\therefore a = b = 0.5$

Now, $\frac{P_B}{P_C} = \frac{2K_1}{K_2} = \frac{2a}{b} = \frac{2x}{y} \Rightarrow \frac{K_1}{K_2} = 1 = \frac{x}{y}$

Now, $P_{10\ min} = (1-x-y) + 2x + y = 1.4$

$\Rightarrow x = y = 0.4 \therefore P_A = 1-x-y = 0.2$ atm at t =10 min

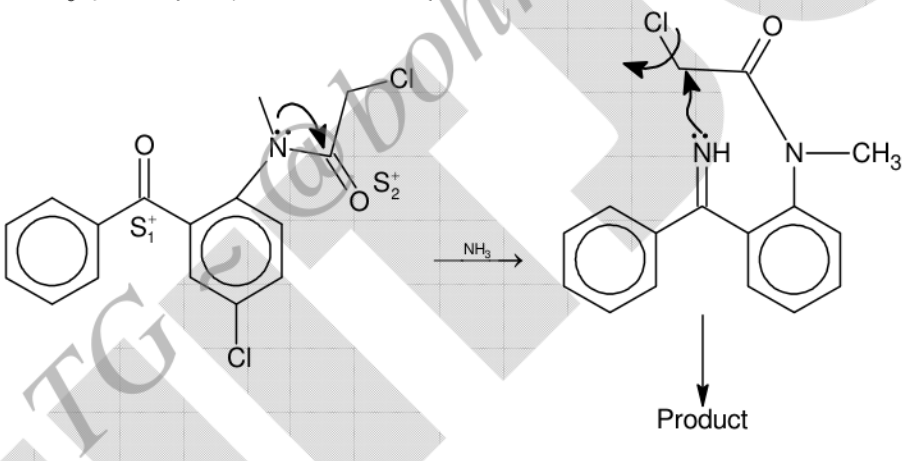
Now, $K_1 + K_2 = \frac{1}{t}.\ln\frac{P_A^o}{P_A} = \frac{1}{10}.\ln\frac{1}{0.2} = 0.16\ min^{-1}$

$\therefore K_1 = K_2 = 0.08\ min^{-1}$

36. C

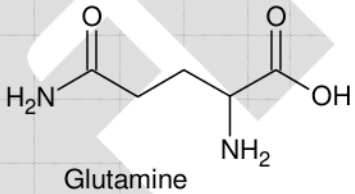
Sol. Rate: Addition of nucleophilic on polar π – bond > Nucleophilic substitution $S_1^+ > S_2^+$

$\therefore NH_3$ (nucleophile) attacks 1st at S_1^+



37. A

Sol.

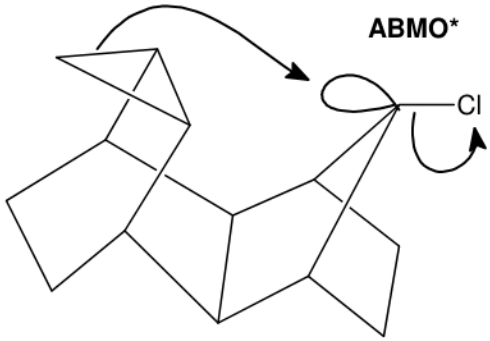


Glutamine

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38. C

Sol.



3 membered ring act as an internal nucleophile.

39. C

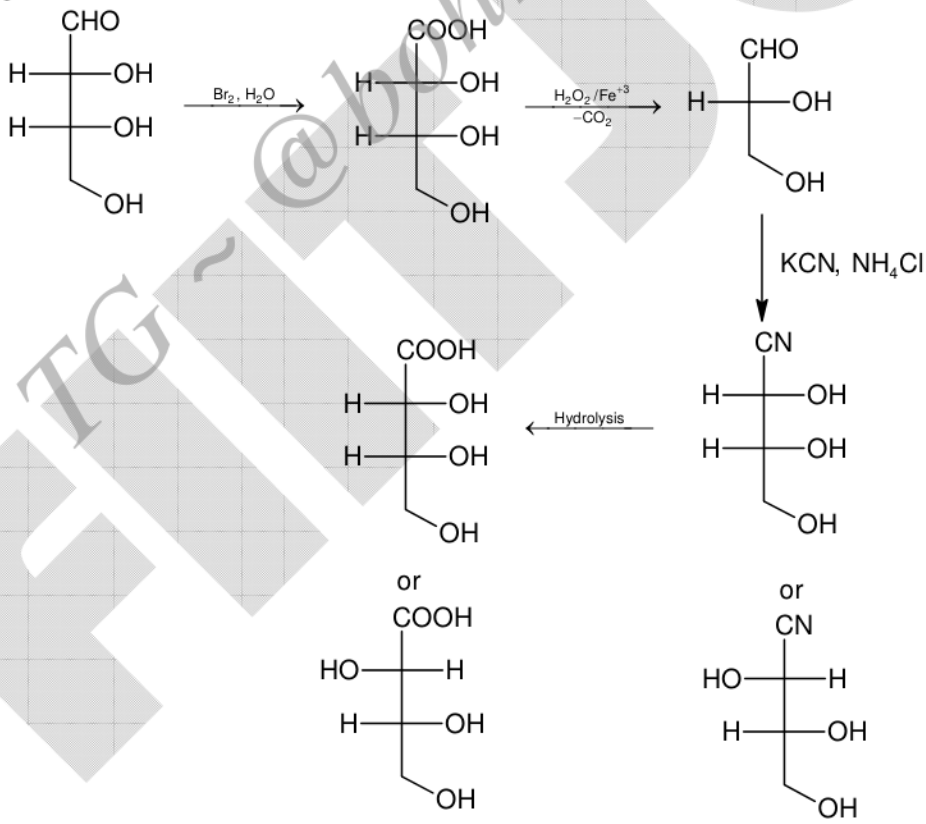
Sol. $H_2C_2O_4$ is obtained in pure form and is not much effected by surrounding conditions and is therefore suitable as a primary standard.

40. C

Sol. Glycerin and H_2O are highly miscible and most water molecules turned to glycerin via H-bond, therefore, vapour pressure of solution decreases. Hence, depression in freezing point is observed.

41. C

Sol.



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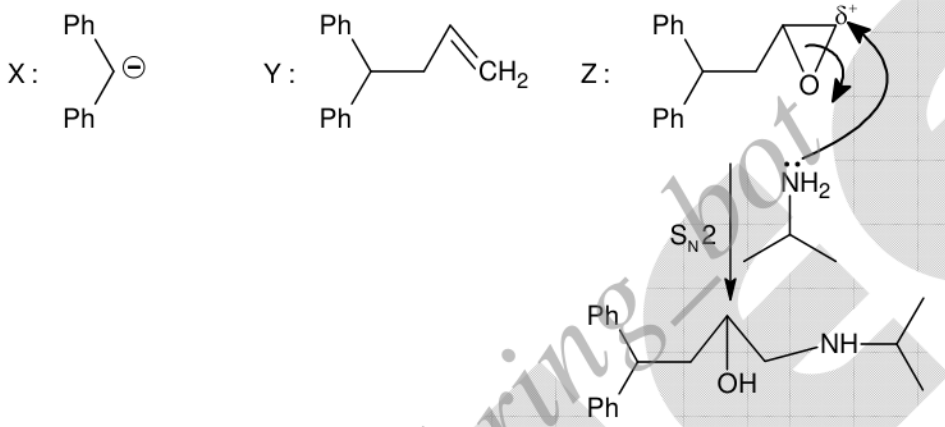
42. B

Sol. $E_{cell} = E^{\circ}_{cell} - \frac{0.06}{n}.\log\frac{\left[H^+\right]^2}{P_{H_2}}$

Or, $0.70 = (0.28 - 0) - \frac{0.06}{2}.\log\frac{\left[H^+\right]^2}{1} \Rightarrow pH = 7.0$

43. B

Sol.



44. B

Sol. Factual

45. D

Sol. Covalent character $\propto \frac{\text{Charge of cation}}{\text{Size of cation}}$

Covalent character $\propto$ size of anion.

SECTION – B

46. 3

Sol. $2x - 8 + 0 + 0 = -2$

$2x = 6$

$x = +3$

47. 6

Sol. $XeF_x + \frac{x}{2}H_2 \longrightarrow Xe + xHF$

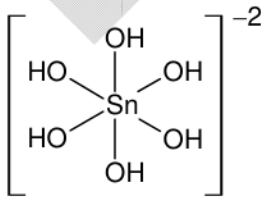
a mole a x mole

$a = \frac{56}{22400}$

and $ax \times 1 = \frac{60 \times 0.25}{1000} \times 1 \Rightarrow x = 6$

48. 2

Sol.



$\rightarrow$ sp^3d^2

$x = 1$

$y = 3$

$z = 2$

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49. 9

Sol. Statement number 1, 3 and 5 are correct.

Statement 3: $Fe^{+3} \longrightarrow \underset{\text{Red-brown}}{Fe(OH)_3}$; $Cr^{+3} \longrightarrow \underset{\text{Green}}{Cr(OH)_3}$

Statement 5: $2CuSO_4 + K_4\left[Fe(CN)_6\right] \longrightarrow \underset{\text{Chocolate brown}}{Cu_2\left[Fe(CN)_6\right]} + K_2SO_4$

50. 7

Sol. At 2nd equivalence point; only $\left[HA\right]^{-2}$ is left.

$$\left[H^+\right] = \sqrt{K_{a_2} \times K_{a_3}} = 10^{-10}$$

Now; $K_{a_1} \times K_{a_2} \times K_{a_3} = \left[H^+\right]^3 \dfrac{\left[A_3^-\right]}{\left[H_3A\right]}$

$$\frac{\left[A^{-3}\right]}{\left[H_3A\right]} = \frac{10^{-3} \times 10^{-8} \times 10^{-12}}{\left[10^{-10}\right]^3} = 10^7$$

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Mathematics

PART – C

SECTION – A

51. D

Sol. $f(0).f\left(\frac{5}{4}\right)$ can be positive, negative or equal to zero.

52. D

Sol. $\theta \in \left[0, \frac{\pi}{2}\right]$

53. C

Sol. $A = I$

54. B

Sol. Centroid, $G \equiv (2, 0)$

Slope of $GH = \frac{1}{3}$

55. A

Sol. Minimum distance between $x = 4\sqrt{1+y^2}$ and $x = -2\sqrt{2y-y^2}$ x is 4.

56. D

Sol. $y^2\cos(x^2y)\,(2xy + x^2y') + x^2y' - 2xy = 0$

$\Rightarrow \quad \cos(x^2y)\,d(x^2y) = d\left(\frac{x^2}{y}\right)$

$\Rightarrow \quad \sin(x^2y) = \frac{x^2}{y} + c$ where $c = 0$

57. B

Sol. $\bar{y} = \frac{\sum_{i=1}^{100} y_i}{100} = \frac{2 \cdot 100 \cdot 101 \cdot 201}{100 \cdot 6} = 6767$

58. A

Sol. $(B, B) \in R \ \forall B \in P(A) \Rightarrow R$ is reflexive

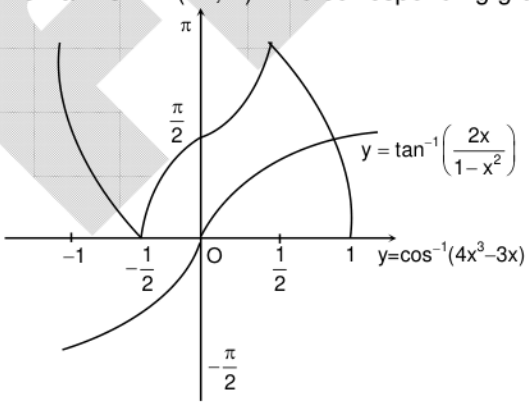
R is not symmetric

$(B, C), (C, D) \in R \Rightarrow (B, D) \in R \quad \forall B, C, D \in P(A)$

So, R is transitive.

59. C

Sol. Domain is $x \in (-1, 1)$. The corresponding graph in this interval is given



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60. A

Sol. $a^3 = 45^3 \Rightarrow a = 45, 45\omega, 45\omega^2$

61. D

Sol. $f(x) = \begin{cases} -\left(\dfrac{e^{x^2-1}-1}{x^2-1}\right) & |x|>1 \\ \sin\left((2x-1)\dfrac{\pi}{2}\right) & |x|<1 \\ 1 & |x|=1 \end{cases}$

62. D

Sol. $\text{Displacement} = \dfrac{3}{2\sqrt{2}}\left[\left(1-\dfrac{1}{\sqrt{2}}\right)\hat{i}+\left(1-\sqrt{\dfrac{3}{2}}\right)\hat{j}\right]$

$\text{time, } t = \dfrac{\frac{3}{\sqrt{2}}\left(\sqrt{\frac{3}{2}}-1\right)}{0.2}$

$\text{Use, velocity} = \dfrac{\text{displacement}}{\text{time}}$

63. D

Sol. $P = \dfrac{\frac{11!}{2!2!2!}-\frac{10!}{2!2!}\times 2+\frac{9!}{2!}}{\frac{9!}{2!2!}{}^{10}C_2} = \dfrac{37}{45}$

64. B

Sol. $\int_0^1 1-\dfrac{x^2}{6}dx < \int_0^1 \dfrac{\sin x}{x}dx < \int_0^1 1-\dfrac{x^2}{6}+\dfrac{x^4}{120}dx$

$\Rightarrow \quad \dfrac{17}{18} < \int_0^1 \dfrac{\sin x}{x}dx < \dfrac{1703}{1800}$

65. D

Sol. $\overrightarrow{AB}-\overrightarrow{BC}+\overrightarrow{CD}-\overrightarrow{DE}+\overrightarrow{EF}-\overrightarrow{FA} = -6\left(\overrightarrow{GH}\right)$

66. A

Sol. $g(xy)+h\left(\dfrac{x}{y}\right) = \dfrac{\left[(xy)^2+xy\right]-\left[(xy)^2-xy\right]}{2}+\dfrac{\left(\frac{x}{y}\right)^2+\frac{x}{y}+\left(\frac{x}{y}\right)^2-\frac{x}{y}}{2} = xy+\left(\dfrac{x}{y}\right)^2$

67. A

Sol. Let $\alpha = x + iy$; $x, y \in I^+$

$\Rightarrow \quad x^2 - y^2 = 4 \Rightarrow x = 2, y = 0$

$\Rightarrow \quad$ slope = 0

68. D

Sol. $\sum_{n=1}^{\infty}\dfrac{1}{(3n-2)!}+\dfrac{1}{(3n-1)!} = e-\dfrac{\left(e+\frac{2\cos\left(\frac{\sqrt{3}}{2}\right)}{\sqrt{e}}\right)}{3} = \dfrac{2}{3}\left(e-\dfrac{1}{\sqrt{e}}\cos\left(\dfrac{\sqrt{3}}{2}\right)\right)$

TG ~ @bohring_bot × @HeyitsyashXD

69. A

Sol. $\alpha = \frac{7}{2}, \beta = -\frac{27}{2}$

70. C

Sol. The circles have 3 common tangents of lengths $2\sqrt{2}$, $2\sqrt{2}$ and 0 respectively.

SECTION – B

71. 9036

Sol. ${}^5C_1({}^{10}C_2 + {}^{10}C_3 + ... + {}^{10}C_5) + {}^5C_2({}^{10}C_1 + ... + {}^{10}C_4) + {}^5C_3({}^{10}C_0 + ... + {}^{10}C_3) + {}^5C_4({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2) + {}^5C_5({}^{10}C_0 + {}^{10}C_1) = 9036$

72. 7

Sol. Let $\tan x = t \Rightarrow \alpha t^2 + 4t(-3 - \alpha) + 64 = 0$

$D < 0 \Rightarrow \alpha \in (1, 9)$.

73. 7

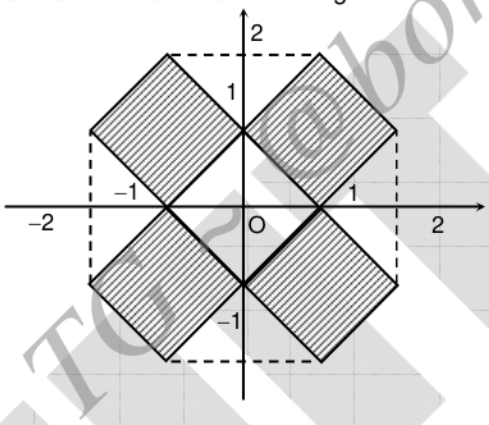
Sol. $I(x) = \frac{1}{2}\ln(xe^x + 1) - \ln(xe^x) + \frac{1}{2}\ln(xe^x - 1) + c$

$= \frac{1}{4}\ln\left(\frac{x^4e^{4x} - 2x^2e^{2x} + 1}{x^4e^{4x}}\right) + c$

$= \alpha = 4, \beta = 2, \gamma = 1$

74. 8

Sol. Set S bounds the shaded region



75. 50

Sol. Coeff. of x^{20} in $(1 - x^2)^{30} = {}^{30}C_{10}$ or ${}^{30}C_{20}$

FIITJEE